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PALEOMAGNETISM OF PROTEROZOIC AND CAMBRIAN STRATA, ADRAR DE MAURITANIE, CRATONIC WEST AFRICA

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Abstract. Four paleomagnetic poles are reported from red beds of the Adrar de Mauritanie, cratonic West Africa. Thirteen localities were occupied, and 224 samples were collected in seven units from a middle Proterozoic to early Paleozoic sedimentary section. Two poles are reported from middle and late Proterozoic strata; a third pole is reported from Cambrian strata. The magnetizations for these reside in stable high-temperature components, revealed by progressive thermal demagnetization analysis to 675°C. The fourth pole is interpreted to reflect a Permian overprinting in the Cambrian strata. Two Cambrian-Ordovician units sampled contained directions too scattered to allow the calculation of any pole. The pole from middle Proterozoic strata (998 ± 32 Ma; Rb-Sr date on argillite), derived from four localities (52 samples, single polarity), plots at 48.7°N, 206.9°E (dp = 1.1°, dm = 2.9°). The pole from late Proterozoic strata (775 ± 52 Ma; Rb-Sr date), derived from two localities (15 samples, mixed polarity), plots at 34.0°N, 246.9°E (dp = 4.6°, dm = 8.6°). Neither has correlatives in age in the paleomagnetic data set for cratonic West Africa, nor are there obvious correlatives with the paleomagnetic data set from the remainder of Africa. The pole that we consider to reflect a Cambrian direction comes from three localities (23 samples, mixed polarity), plotting at 0.3°S, 249.3°E (dp = 5.7°, dm = 11.5°). Its antiparallel pole corresponds with one group of Cambrian poles reported from Africa but differs markedly from another group of Cambrian poles. The late Paleozoic overprint direction obtained from three units (50 samples from four localities, single polarity) provides a (reversed polarity) pole that plots at 47.2°S, 42.6°E (dp = 1.5°, dm = 3.0°), which is in good agreement with Permian poles reported from elsewhere in Africa. In spite of the large number of poles already reported from Africa, additional data are needed before a reliable apparent polar wandering path can be constructed for the Paleozoic and Proterozoic of Africa.

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Introduction

Of the few paleomagnetic studies in Proterozoic and Paleozoic strata of West Africa, two have been carried out in Mauritania. Morris and Carmichael [1978] reported preliminary results from some of the same rocks of the Adrar de Mauritanie that are reported on here (units I2, CO8, and CO10). Because of the limited number of sites, Morris and Carmichael [1978, p. 253] noted that only poles from unit I2 could be used for regional tectonic analysis; these authors also remarked on the deficiency in temporal sequence of the late Precambrian paleomagnetic record for Africa. In the other study, Kent et al. [1984] sampled a unit that correlates with the highest unit studied for this report (CO10), which they investigated about 300 km south of our field area of Atar. They isolated a high-temperature component in some samples but found the directions to be randomly distributed.

Paleomagnetic data for Africa have not yet been able to provide arguments for or against the existence of a Pan African ocean in late Proterozoic time. Because of this, a new systematic paleomagnetic study of the flat lying sedimentary section of the Adrar de Mauritanie seemed needed. The section ranges from middle Proterozoic (~ 1 Ga) to Late Devonian in age, contains a number of horizons of fine-grained red beds, and includes the time of the Pan African orogeny (~ 600 Ma).

Location and Setting

The West African craton consists of a 2.5-2.7 Ga metamorphic and granitic basement that stabilized at about 1.7 Ga. Proterozoic strata began to accumulate on this craton at about 1 Ga, with sedimentation continuing across late Proterozoic and into Paleozoic time [Sougy, 1973]. Orogenic belts of two discrete ages enclose the stable craton (Figure 1a). The older belt is related to the Pan African orogeny of late Proterozoic age (about 600 Ma [Black, 1984; Liegeois and Black, 1984; Caby et al., 1985; Caby, 1986]). Rocks deformed at this time are exposed in a number of areas around the old craton. These include South Morocco, Hoggar and Adrar des Iforas, Gourma, and the Dahomeyides and Rokelides ranges. The younger orogenic belt on the west and northwest is thought to be of Caledonian and Hercynian age [Lecorche and Sougy, 1978; Lecorche, 1980]. It includes the Mauritanides Range [Sougy, 1969], which immediately borders the study area on the southwest [Trompette, 1973].

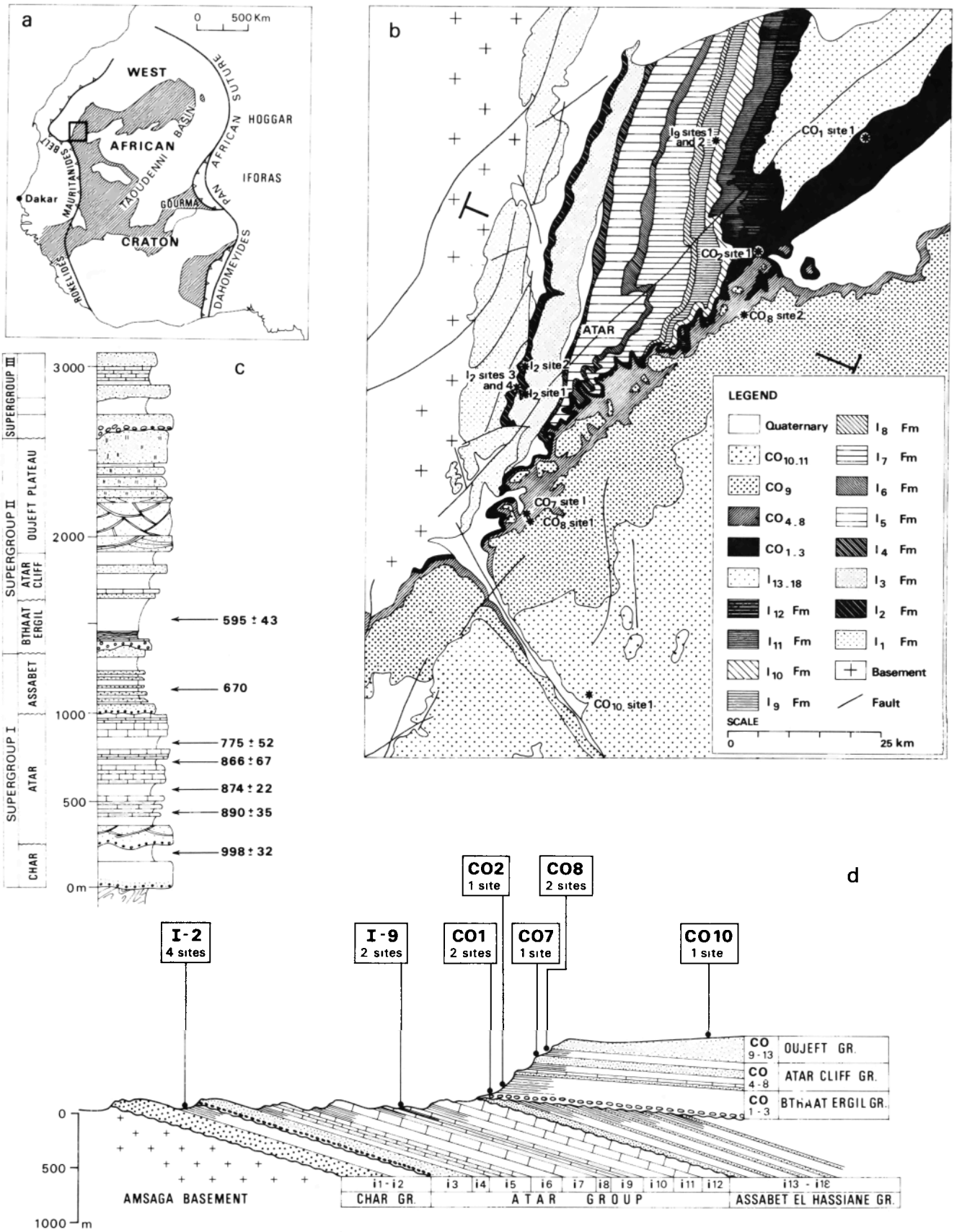


Fig. 1. (a) Index map of West Africa showing location of study area and structural boundaries of West African craton. (b) Geological sketch map of Adrar area, Adrar de Mauritanie (after Trompette, 1973) showing sample localities. (c) Generalized stratigraphic section of sedimentary strata of the Adrar de Mauritanie [after Trompette, 1973]. (d) Schematic cross section (line of section indicated on Figure 1b).

TABLE 1. Site Location Data, Adar de Mauritanie

Site Identification		Site Location		Bedding		Number of Samples	
Strata	Site	Latitude °N	Longitude °E	Strike, deg	Dip, deg	Cores	Blocks
I2	1	20.44	346.86	355	5 E	19	--
I2	2	20.48	346.87	350	4.5 E	24	15
I2	3	20.44	346.86	352	3.5 E	9	--
				10	4 E	6	--
				355	1 E	7	--
I2	4	20.44	346.86	340	1.5 E	33	--
I9	1	20.81	347.16	34	8 E	--	8
I9	2	20.81	347.16	--	--	11	--
CO1	1	20.76	347.38	--	--	15	--
CO2	1	20.64	347.24	258	4.5 N	--	8
CO2	2	20.63	347.24	330	3 N	--	6
CO7	1	20.25	346.89	203	6 W	4	--
				252	2 N	8	--
CO8	1	20.24	346.89	105	3 S	24	--
CO8	2	20.54	347.21	98	3 S	12	1
CO10	1	20.01	346.93	240	3 N	15	--

The Adrar de Mauritanie (Figures 1a and 1b) is within the structurally simple Taoudeni basin, one of the largest Precambrian and Paleozoic sedimentary basins of Africa. The sedimentary filling of the basin has been well described by Trompette [1973] in the area of the Adrar de Mauritanie (Figure 1b). The stratigraphic section is subdivided into three supergroups (Figure 1c). We are here concerned only with supergroups 1 and 2.

Supergroup 1 at the base of the section, about 1.4 km thick, unconformably overlies the Amsaga basement. It has been assigned a late Precambrian (middle and late Proterozoic) age, derived from (1) Rb-Sr sedimentary rock age determinations on illitic fine fractions [Clauer, 1976; Clauer et al., 1982] (Figure 1c), (2) the paleontologic study of stromatolites [Bertrand-Sarfati, 1972] and microfossils [Amard, 1986], and (3) the correlation of continental tillites overlying this supergroup with a late Proterozoic glacial event [Trompette, 1973; Deynoux, 1978; Deynoux and Trompette, 1981]. Supergroup 1 has been correlated in detail on stratigraphic grounds with Proterozoic sequences elsewhere in West Africa [Bertrand-Sarfati, 1972; Trompette, 1973]. Supergroup 1 is divided into three groups that are separated by unconformities. The lower group (Char Group) consists of coarse- to fine-grained red beds. The overlying Atar Group is characterized by interbedded carbonate and fine clastic deposits that are dominantly gray. The upper group (Assabet Group) consists of fine clastics.

Supergroup 2, about 1.2 km thick, consists of dominantly gray, fine clastic and subordinate carbonate deposits. An irregular disconformity marks the base, which is directly overlain by a tillite that is considered to mark the top of the Proterozoic elsewhere in West Africa [Deynoux, 1978]. An Rb-Sr isotopic study on sedimentary rocks of unit CO3 near the base of supergroup 2 has provided an age of 595 ± 43 Ma [Clauer, 1976]. Two species of inarticulate brachiopods near the top of supergroup 2 (units CO12 and CO13, Figure 1d) indicate accumulation near the Cambrian-Ordovician boundary [Legrand, 1969].

A total of 224 oriented samples were collected from hematite-bearing strata at 13 localities from seven units that span parts of supergroups 1 and 2. Of these, 186 cores (2.5 cm in diameter and about 3-7 cm in length) were obtained using a portable diamond core drill similar to equipment described by Doell and Cox [1967]. Thirty-eight block samples also were collected. Sampling localities are shown in Figure 1b. Stratigraphic positions of the intervals sampled are depicted in Figure 1d. Table 1 summarizes information as to location, tilt correction, and the number of samples collected. Because of limited exposures, only a few hematite-bearing samples could be collected at some localities. Stratigraphic descriptions of the localities, and of equivalent strata, are given by Trompette [1973]. The cores and block samples were oriented by magnetic compass, read to the nearest half degree. Only small, essentially negligible corrections for tilt were required (Table 1).

Macroscopic and Microscopic Observations

The timing of acquisition of stable magnetizations in ancient red beds and the utility of ancient red beds for paleomagnetism have been the subject of some discussion. For contrasting views, see (1) Elston and Purucker [1979], Purucker et al. [1980], Bressler and Elston [1980], and (2), Walker et al. [1981], Larson et al. [1982], and Larson and Walker [1982]. The second group of workers contends that ancient (Mesozoic and older) red beds accumulated as gray, magnetite-bearing sediments and that their alteration and reddening occurred some geologically appreciable time after deposition, compromising their utility for paleomagnetism and magnetostratigraphy. This model for red bed formation is argued principally from the observation that altered detrital grains exist in the red beds (to which the magnetization has not

been related) and from the extrapolation of field relations reported from partially reddened sediments inferred to be modern analogues of the ancient red beds. In contrast, we maintain that general and detailed field stratigraphic relations, stratigraphically reproducible antiparallel reversals and reversal boundaries, and reproducible paleomagnetic directions obtained on regional and local scales form a coherent body of evidence indicating that ancient red beds accumulated as red beds and that their characteristic magnetizations were acquired during the course of deposition, lithification, and early diagenesis.

From detailed studies aimed at identifying the carriers of magnetization and the time(s) of acquisition of the early stable magnetizations in ancient red beds, it has been noted (for example, in our studies listed above) that the paleomagnetic directions reside partly in red hematite pigment (Fe_2O_3) related to an early chemical magnetization and partly in detrital grains of black hematite (specularite; also Fe_2O_3). From demagnetization analysis, the hardest magnetizations reside in the specularite. Moreover, contact relations between grains in heavy mineral laminations indicate that the specularite became hematite in the source terrain (apparently mainly from the alteration of magnetite) and that partial to extensive alteration of some of the detrital specularite grains to red hematite also must have occurred in the source terrain and during transport to the place of deposition. Additionally, the relatively large detrital specularite grains appear to have been the source for the ubiquitous, tiny angular grains of specularite disseminated throughout the red beds, apparently arising from the mechanical disintegration of coarser specularite grains during transport. Last, a demonstrable bedding error residing in specularite has been recognized in normally and reversely magnetized strata from the Lower Triassic Moenkopi Formation of the Colorado Plateau [Elston and Purucker, 1979]. The existence of these bedding errors argues for an early formation of the specularite and an early acquisition of the stable magnetization on grounds that are independent of other supporting field geologic relations.

We here favor the model that calls for formation of ancient red beds during the course of deposition and lithification, accompanied by contemporaneously acquired directions of magnetization residing in detrital specularite and an early formed hematite pigment. However, we also recognize that many ancient red beds have undergone minor to extensive postdepositional alterations, accompanied by the precipitation of secondary hematite (commonly pigment, and in some cases secondary specularite), and where this has occurred, the rock contains secondary as well as earlier directions of magnetization. Where ancient red beds have undergone secondary hematizations, evidence commonly (but not always) is recognizable at the outcrop. In such rocks, original directions of magnetization commonly are masked by the secondary directions of magnetization. In many cases, the secondary magnetizations cannot be entirely removed by any cleaning technique, and spurious directions of magnetization result even though progressive demagnetization analysis seems to reveal univectorial decay and only a single component of magnetization. We thus conclude that the key to any successful paleomagnetic study of ancient red beds hinges on the sampling of strata that have undergone very little or no postdepositional alteration and, alternatively, where the strata do contain secondary magnetizations, on the discrimination of early directions from secondary directions of magnetization. The latter objective has proven difficult to attain.

Strata sampled for this study exhibit textural and mineralogical characteristics and paragenetic relations between the hematite and host matrix material, typical of ancient red beds. Demagnetization analysis indicates that the remanent magnetization resides in hematite, in accord with microscopic observations under combinations of reflected and transmitted light. Under the microscope, hematite is seen to exist in two forms: (1) opaque grains of black hematite (specularite) that form the major part of a population of detrital heavy minerals and (2) finely disseminated hematite pigment that imparts the red color to the rock. In several of the units sampled, the hematite exhibits interrelationships that argue for an early reddening of the rock. It is a reddening not obviously related to the formation or alteration of the detrital grains of specularite. In some other units, however, relations argue for a secondary, postlithification origin for some or most of the hematite, suggesting the existence of secondary components of magnetization. Rocks containing observable secondary mineralizations correlate with secondary magnetizations revealed by progressive thermal demagnetization analysis. In contrast, siltstone, mudstone, and sandstone strata that display little or no evidence for postdepositional modification or emplacement of red and black hematite contain magnetizations that we consider to be early and to have been acquired during and shortly after accumulation and lithification of the beds as a consequence of detrital and early chemical processes.

Characteristics observed in the sampled units are briefly summarized below. In apparently unmodified red beds the strata, although made up of altered materials, are considered to be "unaltered." In red beds containing obvious secondary mineralizations, the strata are considered to be altered from a previously "unaltered" condition. At all of the localities sampled for this study, the red beds exhibited at least some secondary magnetizations reflecting at least minor, physically unidentifiable alterations. These samples responded to thermal demagnetization analysis at the lower temperatures, indicating that such easily removable secondary magnetizations reside in pigment rather than specularite.

Hematite in Unaltered Red Beds

Pigment. Units I2, I9, CO7, and CO8 consist of purple-red to red-brown siltstone, mudstone, and sandstone. Within many individual beds, small to tiny siltstone and mudstone clasts derived from adjacent beds have colors and textures distinct from the enclosing matrix, attesting to an apparent predisruption reddening of the fragments. Additionally, very fine grained beds that exhibit intense reddening with respect to adjacent coarser clastic strata locally show the effects of soft sediment deformation, which also

attests to an early development of hematite pigment. The rip-up clasts and deformation features were too small to be sampled for conglomerate or fold tests. In general, the distribution of pigment in the individual beds is not relatable to the distribution and alteration of heavy mineral grains. Rather, the pigment in each bed appears to simply reflect the existence of differing amounts of red hematite available at the time of deposition and lithification.

Unaltered red beds characteristically display fine disseminations of nonred spots, seen as a bleachlike spotting or freckling in the siltstone, mudstone, and sandstone. The freckles are distributed in a manner that appears to record an incomplete impregnation of the sediment by hematite pigment at the time of deposition rather than a postdepositional leaching arising from a later invasion of solutions. Some nonred spots appear deficient in disseminated grains of specularite as well as pigment, but other non-red spots contain disseminated specularite. In either case, the grains of specularite do not appear to have been the source for, or to have contributed to the production of, the disseminated red pigment.

Specularite. Heavy mineral grains consisting of specularite and subordinate nonopaque heavy mineral grains are disseminated in most of the sampled red beds. In sandstone the grains may exhibit partial, and very rarely complete, rounded detrital outlines. In the finer-grained rocks, however, most specularite grains exhibit highly angular shapes, and partially rounded detrital outlines are subordinate to rare. The differing physical characteristics do not really pose a problem. Tiny angular fragments of specularite are found disseminated in the sandstone as well as in the finer-grained rocks, and specularite grains having partial detrital outlines also are found in the slightly coarser fractions of mudstones, in proximity to claystone and the finer grained siltstone beds that contain only angular grains of specularite. The angular specularite grains are most simply explained as fragments derived from the breakup during transport of larger (silt-sized and very fine) grains of detrital specularite. The finest specularite grains simply have not undergone sufficient abrasion in the fine-grained environments of deposition to have reacquired detrital outlines, nor does the formation of rounded, detrital outlines appear to have occurred in the tiny grains regardless of the energy level of the environment of deposition.

Of importance is the observation that both the coarse and fine specularite grains in unaltered red beds exhibit crisp contacts with adjacent nonopaque and opaque grains. There characteristically are no crusts or rinds of secondary hematite on the specularite grains, nor do halos of red hematite enclose these grains as would be expected if the specularite had formed from the in situ alteration of magnetite. Rather, the differing degrees of alteration to red hematite observed in different grains are restricted to the individual grains, even when the grains are in grain-to-grain contact in heavy mineral laminations (such as described by Elston and Purucker [1979]). Such paragenetic relations, which are analagous to and which parallel relationships seen in beds and clasts in the field, indicate that the specularite formed prior to deposition and not as the consequence of postdepositional in situ alterations.

In the Mauritanian red beds, heavy mineral grains of different origins locally are found in heavy mineral laminations. The various grains consist of (1) grains of magnetite-ilmenite that have been converted to the martite variety of specularite, (2) grains displaying intergrowths of hematite and ilmenite indicative of an igneous origin, (3) occasional grains of zircon, and (4) a grain of chromite (identified under the electron microprobe by J. Clark, Arizona State University). Paragenetic relations in the heavy mineral laminations, similar to those summarized by Elston and Purucker [1979], alone are strong evidence for predepositional formation of the specularite in the source terrane (such as described by Eggler et al. [1969, p. 511]) and for the alteration of grains in the source terrain and during transport to the place of accumulation.

Postlithification Hematite

Veinlets of a secondary, very finely crystalline hematite that both parallel and crosscut the bedding are present in some of the sampled units. Veinlets observed in locality 3 of unit I2 are in rock containing a secondary magnetic overprint that could not be completely removed by thermal cleaning. Veinlets and an irregularly distributed secondary pigment also have been observed in unit CO2. These features are associated with a deeply pigmented rock that lacks "freckles" and that contains no specularite of apparent detrital origin. We conclude that unit CO2 has undergone a general secondary mineralization and remagnetization. The stable magnetization obtained from unit CO2 has provided data for the calculation of an overprint direction and pole.

Unit CO1 contains an anomalous direction suggestive of overprinting. Although disseminated specularite is present, the pigment is irregularly distributed, and the specularite grains are trapped in a mozaic of quartz grains that bears evidence for extensive postdepositional dissolution of silica.

Last, unit CO10 displays abundant evidence at the outcrop for the secondary mobilization and precipitation of hematite (abundant laminations of hematite pigment crosscut the stratification in this unit). Unit CO10 also exhibits evidence for extensive alteration of the rock on a microscopic scale. The quartz grains are deeply sutured and interlocked, which is evidence for extensive postdepositional solution of silica and a dissolution that almost certainly affected the distribution of an original hematite pigment and specularite. Hematite pigment is sparse and irregularly disseminated, a silt and clay matrix that could contain tiny grains of specularite is lacking, and only coarse specularite grains (many deeply altered) are disseminated in the rock. Paleomagnetic analysis of this rock has shown only highly scattered but extremely stable directions of magnetization, which presumably reside in the coarse specularite. Because obvious overgrowths on the specularite grains are lacking, the scattered directions conceivably may reflect rotations of grains accompanying dissolution of silica.

Magnetic Methods

Laboratory measurements were carried out at the Paleomagnetics Laboratory of the U.S. Geological

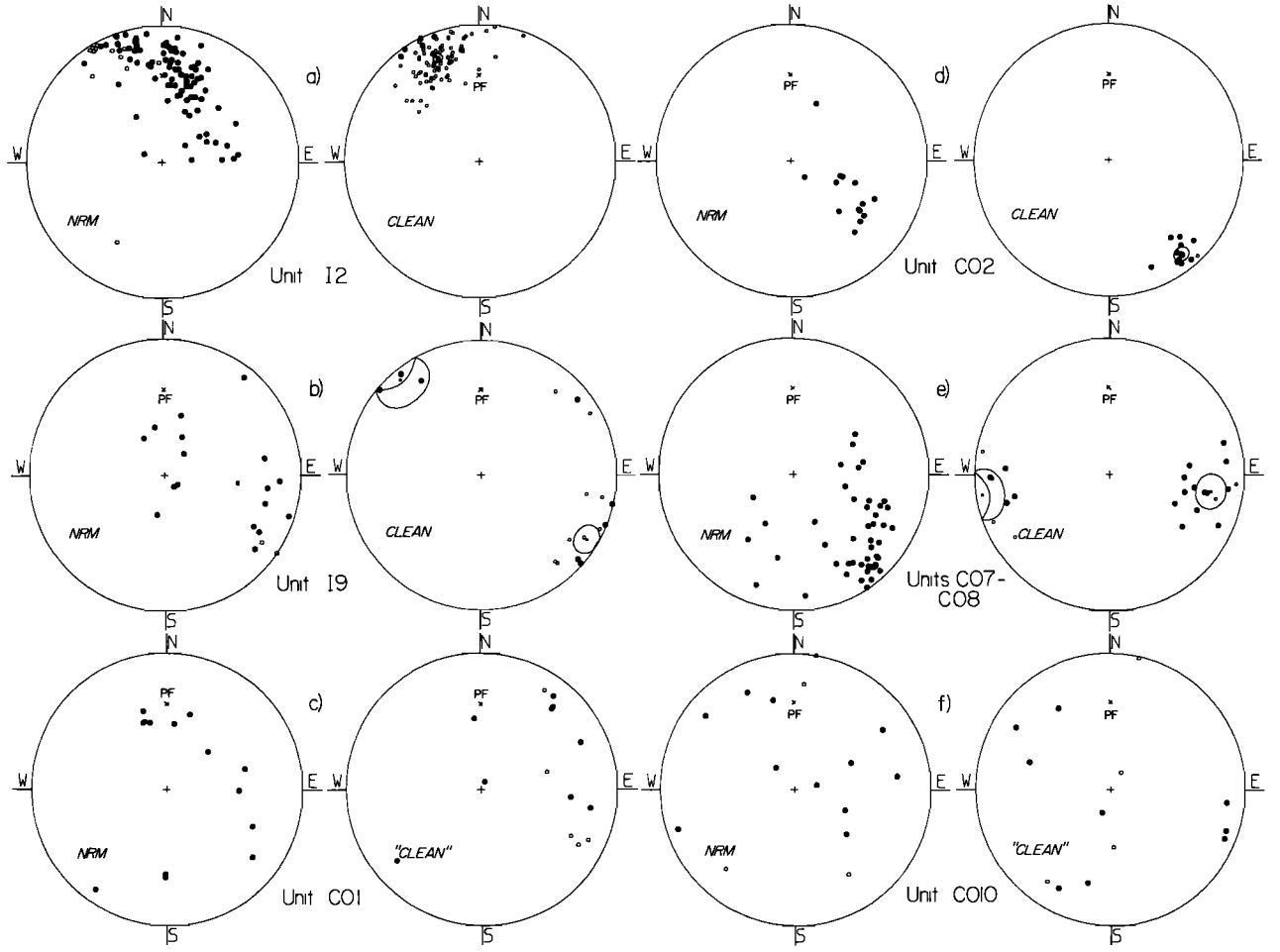


Fig. 2. Equal-area plots of directions (open (solid) symbols indicate upper (lower) hemisphere) of natural remanent magnetization (NRM) and mean directions following cleaning, by stratigraphic units, Adrar de Mauritanie.

Survey at Flagstaff, Arizona. One specimen obtained from each core was subjected to progressive thermal demagnetization analysis. End pieces of cores were employed for chemical demagnetization analysis. Prior to demagnetization analysis, samples were stored in a low field for 3 weeks to allow possible components of a viscous remanent magnetization (VRM) to decay. Measurement of remanence was carried out on a cryogenic magnetometer of commercial manufacture, coupled to a microcomputer. Noise level of the magnetometer is of the order of 10^{-7} emu (10^{-10} Am^2). Alternating field (af) demagnetization employed a tumbling af demagnetizer of commercial manufacture. The samples were thermally demagnetized in a noncommercial, noninductively wound, mu-metal shielded oven having a residual field of less than 5 gammas and a thermal gradient in the heating region of ~5°C. Pilot samples from each locality were first subjected to alternating field demagnetization analysis and then either to chemical or thermal demagnetization analysis.

The af demagnetization of pilot samples in fields up to 100 mT failed to affect significantly the various components of magnetization, and only small changes in direction and intensity were observed. Magnetite therefore is not an important carrier of magnetization in these ancient red beds. Nonetheless, all samples were cleaned in 10-mT alternating fields prior to thermal demagnetization analysis. This precautionary step was carried out to remove the effects of secondary magnetizations having low coercive forces but potentially high blocking temperatures, such as piezoremanent magnetizations (PRM) potentially imposed by pressure (e.g., lithostatic, tectonic) and possibly some viscous remanent magnetizations (VRM).

As reported by others [e.g., Park, 1970], the magnetization in ancient red beds resides in some combination of hematite pigment (red hematite) and specularite grains (black hematite). The strategy in chemical demagnetization is to remove the pigment and its magnetization selectively with respect to magnetization residing in grains of detrital(?) specularite. Chemical demagnetization was carried out on end pieces of cores from all representative units. Minimum porosities [Henry, 1979] for the samples ranged from 2 to 16%. The samples were immersed in individual beakers each containing 20 cm^3 of HCl (6 N). Measurements were made after 2, 20, 40, 90, 140, 210, 300, 400, 500, 640, 800, and 1150 hours. The results of the chemical demagnetization analysis were less than wholly satisfactory. First, the strength of the acid had to be reduced from a commonly used 10-12N [Burek, 1969; Henry, 1979; Collinson, 1983] to 6N in order to prevent a rapid and thorough

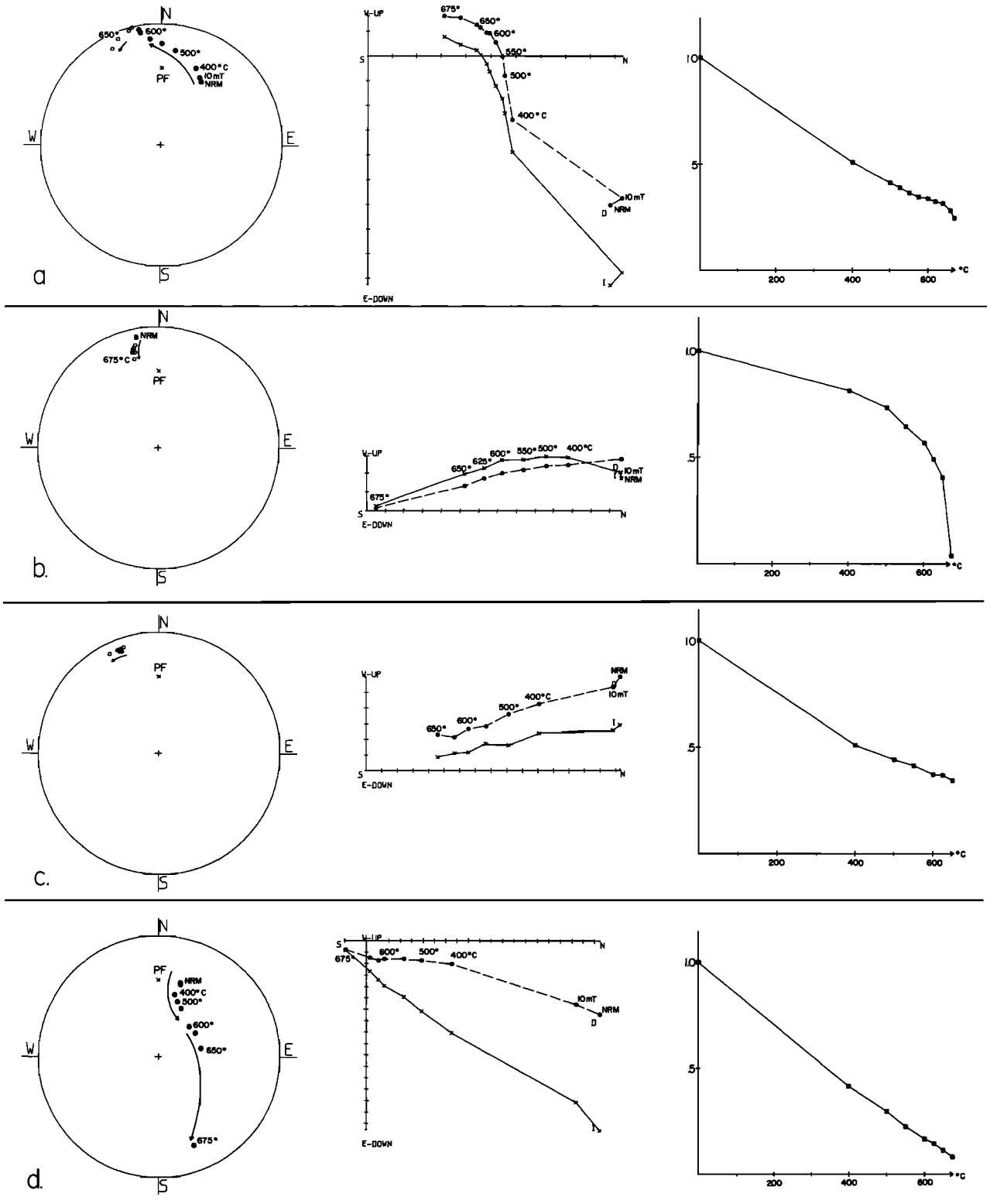


Fig. 3. Equal-area plots of directions, orthogonal demagnetization diagrams, and normalized plots of intensity for progressive thermal demagnetization analysis of representative samples. Arrows indicate direction of vector movement on stepwise cleaning.

dissolution of the samples, which occurred in less than 70 hours in the strong acid. Even in the weaker acid, both the red and black hematite of some samples appear to have been dissolved concurrently, and this was characterized by an erratic evolution of the demagnetization record. In other samples, however, pigment was removed before substantial dissolution of the specularite occurred. In such cases, directions observed after chemical cleaning generally coincided with results obtained from stepwise thermal demagnetization analysis. Because the efficiency and

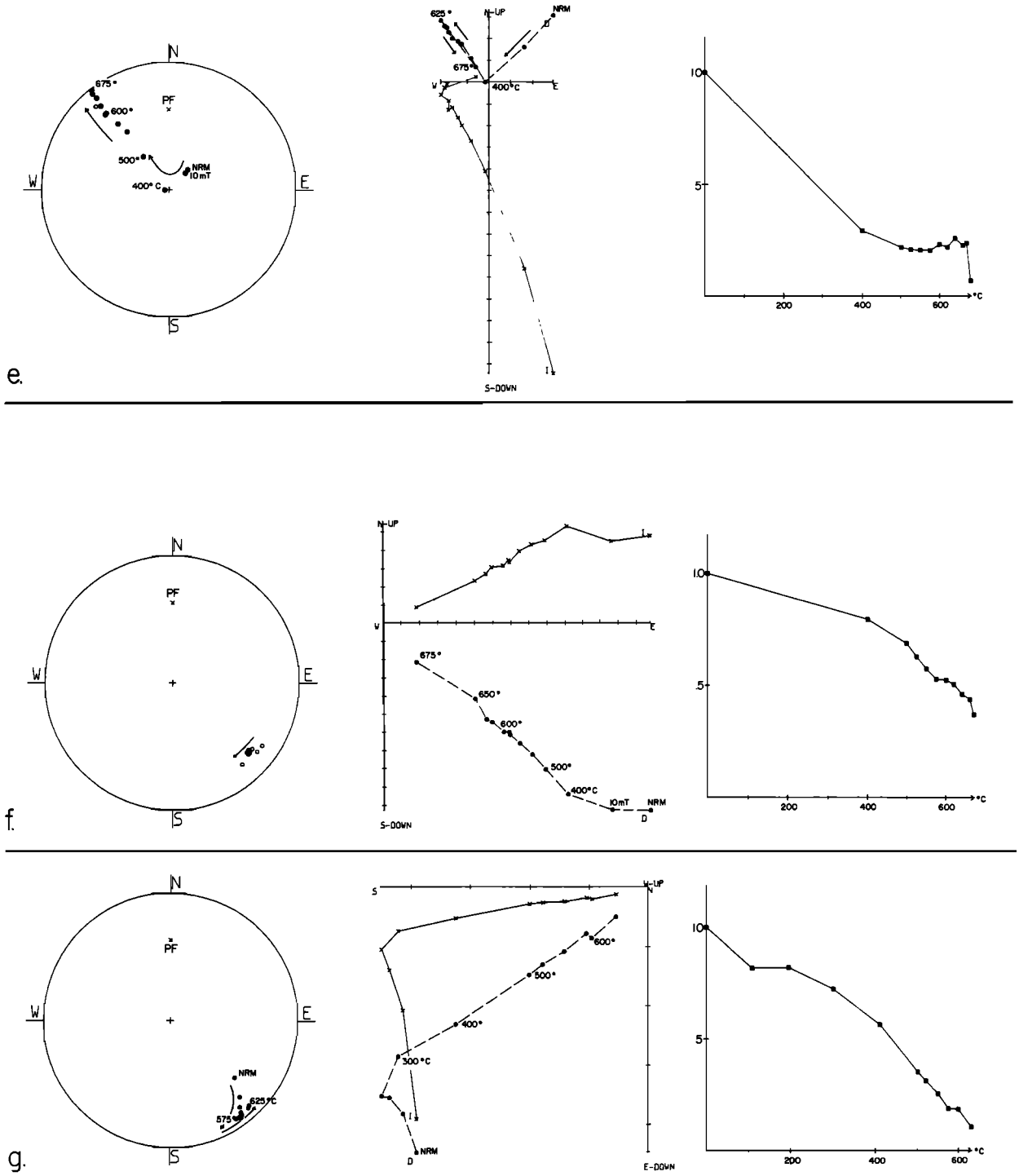


Fig. 3. (continued)

efficacy of chemical cleaning were considerably poorer than that of the thermal demagnetization, we did not further pursue this part of the study, and we do not report any other results from the pilot samples.

Progressive thermal demagnetization analysis was carried out on the remaining samples. Initially, 48 pilot samples were demagnetized at 100°C steps for the interval from 200°C to 500°C and at 25°C intervals from above 500°C to 675°C. Because only small changes in direction and intensity were observed below 500°C, the remaining samples were demagnetized at steps of 400°, 500°, 550°, 600°, 625°, 650°, and 675°C.

Results

The directions of natural remanent magnetization (NRM) are depicted in Figure 2 by stratigraphic units. Intensities of the red beds are characteristically weak, ranging from approximately 10^{-5} to 10^{-7} emu/g (10^{-5} to 10^{-7} Am^2

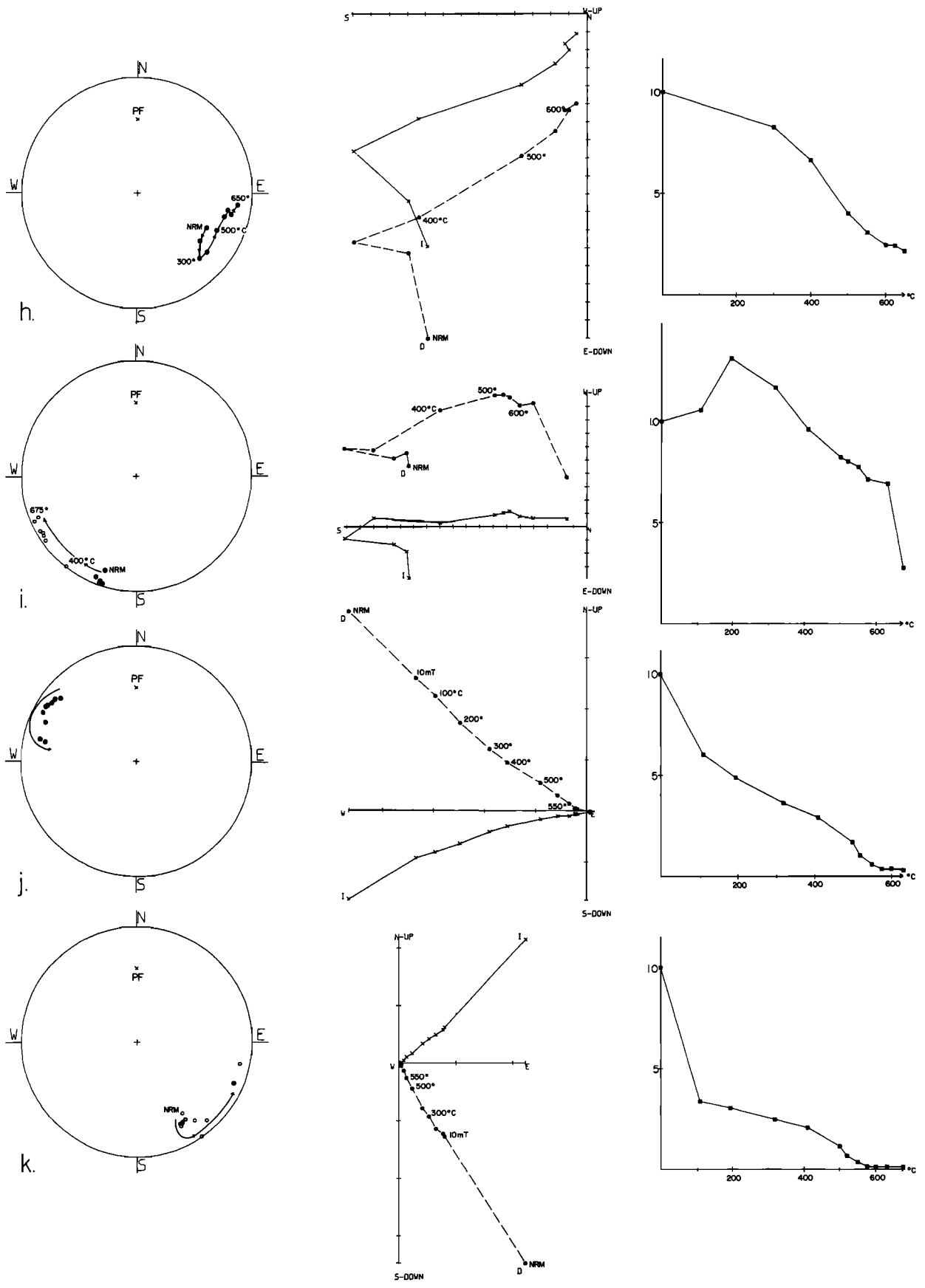


Fig. 3. (continued)

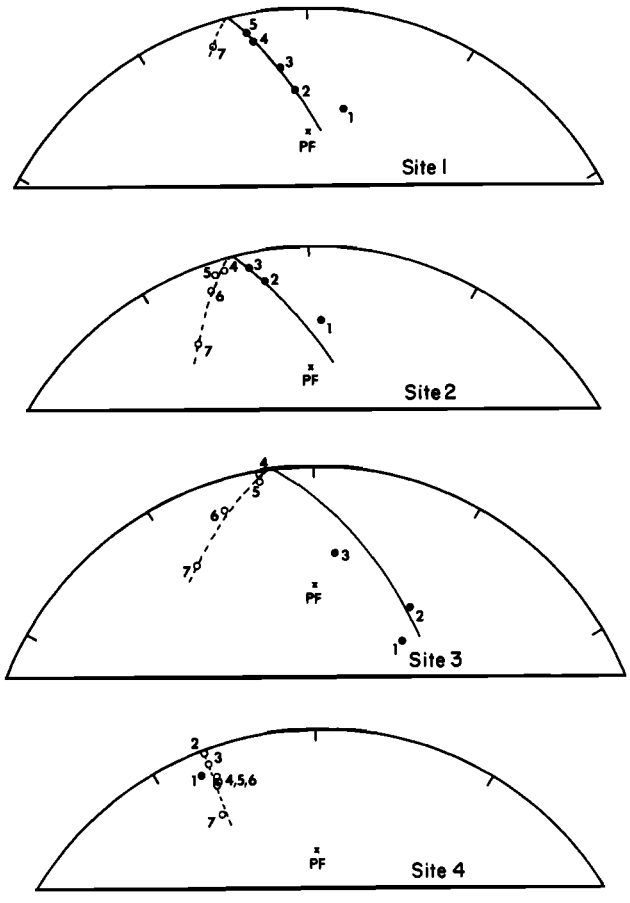


Fig. 4. Equal-area plots for unit I2, showing mean directions for each step of the progressive thermal cleaning, by localities. The demagnetization steps are 1, NRM; 2, 400°C; 3, 500°C; 4, 600°C; 5, 625°C; 6, 650°C; 7, 675°C.

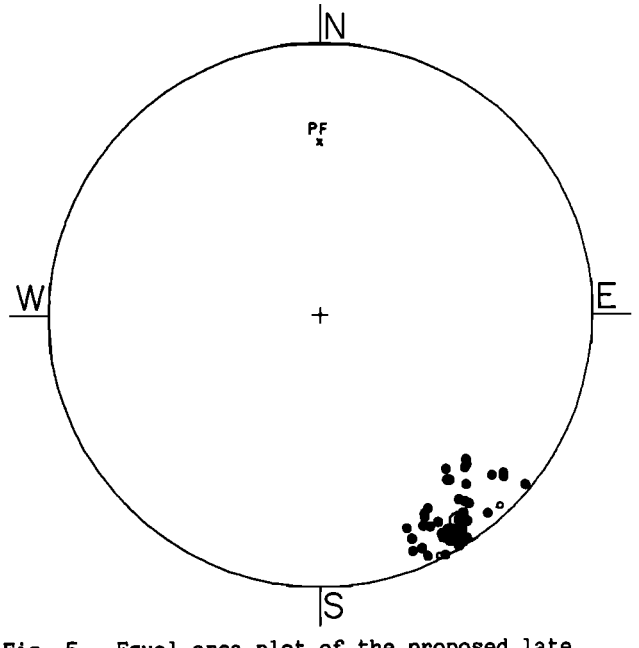


Fig. 5. Equal-area plot of the proposed late Paleozoic overprint direction derived from units CO2, CO7, and CO8, Adrar de Mauritanie.

Kg^{-1}). The NRM directions for most sample groups displayed streaked distributions that reflected the existence of secondary components of magnetization.

Progressive demagnetization data have been evaluated employing standard analytical techniques and projections, which include equal area plots of directions, orthogonal demagnetization diagrams [Zijderveld, 1967; Dunlop, 1979], normalized plots of intensity with progressive demagnetization, and principal component analysis [Kirschvink, 1980] (Figures 2, 3, 4, and 5).

Proterozoic Strata

Unit I2. Most samples from unit I2 displayed multicomponent magnetizations. The demagnetization behavior of the I2 samples, portrayed by locality, is depicted in Figure 4. For these, the site mean directions are shown for each step of the progressive demagnetization analysis. Except

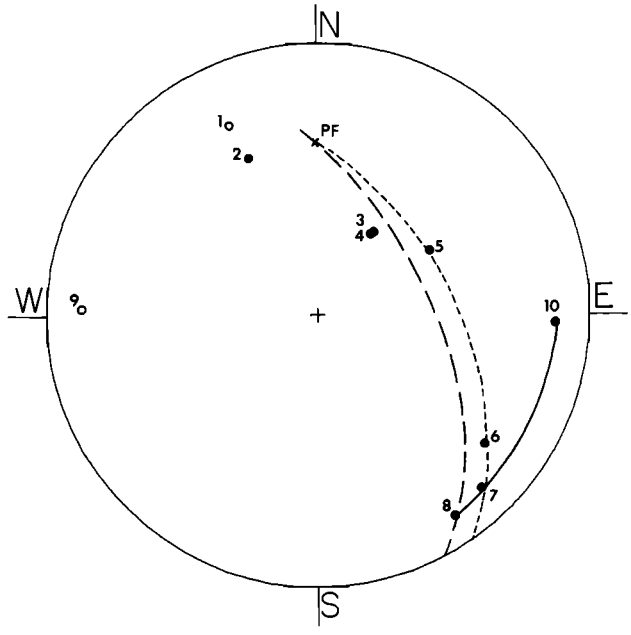


Fig. 6. Comparison of primary and secondary directions of magnetization from Morris and Carmichael [1978], from Kent et al. [1984], and from this study, Mauritania. 1, Clean direction for unit I2 (this study); 2, Morris and Carmichael's secondary component of magnetization for unit I2; 3, Morris and Carmichael's initial component of magnetization for unit I2; 4, Morris and Carmichael's initial component of magnetization for unit CO10; 5, Morris and Carmichael's initial component of magnetization for unit CO8; 6, Morris and Carmichael's secondary component of magnetization for units CO8 and CO10; 7, Kent et al.'s secondary components of magnetization for the Mejeria red sandstone (equivalent to unit CO10); 8, Late Paleozoic overprint direction for units CO2, CO7, and CO8 (this study); 9 and 10, mixed polarity Cambrian direction for units CO7 and CO8 (this study). One great circle (short dashed lines) connects direction 6 with the present field direction; another (long dashed lines) connects direction 8 with the present field direction; and the third (solid) line connects directions 8 and 10.

TABLE 2. Mean Paleomagnetic Directions and Poles, Adrar de Mauritanie

		Clean Directions						VGP				
Units	Age, Ma	B(N)	D, deg	I, deg	k	a_{95}	S-P	Latitude, deg	Longitude, deg	dp	dm	Plat
I2	998±32	4(17)	334.9	-23.5	59	4.7	P	49.1	206.1	2.9	5.2	
		4(52)	334.4	-23.6	63	2.5	P	48.7	206.6	1.1	2.9	-12.3
		4(89)	337.9	-19.6	23	3.2	P	52.6	204.5	1.7	3.3	-9.9
I9	775-866	1(3)	319.8	8.3	43	19.0	P	47.6	239.7	9.7	18.9	4.2
		2(12)	121.5	-9.0	21	9.7	P	-31.0	69.7	5.2	9.7	-4.5
		2(15)	125.2	-6.8	20	8.6	P	-34.0	66.9	4.6	8.6	±3.4
C07-C08	Cambrian	2(8)	260.8	5.5	12	16.8	P	-7.6	262.8	8.6	16.6	2.8
	595	3(15)	99.1	24.6	13	11.0	P	-3.8	61.7	6.3	12.0	12.9
		Average	89.9	9.5	---	---	P	1.7	72.5	---	---	±4.8
		5(23)	92.4	14.4	9	11.0	P	.3	69.3	5.7	11.5	±7.3
C02	Late	2(13)	142.2	13.6	66	5.1	S	-43.8	44.6	2.9	5.2	6.9
C07	Paleozoic	1(9)	140.0	24.5	43	7.1	S	-38.6	40.2	4.0	7.4	12.8
C08-1	(Permian,	1(18)	150.0	4.8	259	2.0	S	-52.9	42.8	1.1	1.7	2.4
C08-2	reversed	1(10)	146.3	10.8	49	7.0	S	-47.9	42.7	3.4	6.9	5.5
LPO	polarity)	4(50)	145.7	11.8	45	3.0	S	-47.2	42.6	1.5	3.0	-6.0

B(N), number of localities (number of samples); D, declination; I, inclination; k, precision parameter of Fisher [1953],n-1/n-r; a_{95}, half angle of cone at 95% confidence level; S-P, component of magnetization interpreted to be secondary or primary; dp, dm, semiaxes of oval at 95% confidence level; plat, paleolatitude.

for the 400°C and 500°C temperature steps of locality 3, a great circle can be drawn that fits the directions obtained from stepwise cleaning of samples from each locality. The great circle is most simply characterized as reflecting two components of magnetization. However, the NRM directions are off of the circle, indicating the existence of one or more, other low-temperature components. Although the evolution of directions for unit I2 during cleaning is homogeneous, the mean "clean" direction is not easily determined. Even after the highest thermal step, the directions remain "streaked," indicating that some unremoved secondary components of magnetization remain. The existence of secondary components also is seen on the orthogonal diagrams. This, indicative of incomplete cleaning (Figure 3a), is a common problem when dealing with red beds. Different methods have been proposed to deal with the skewed directions. They include remagnetization circles [Halls, 1976], mixed remagnetization circles and direct observational data [Bailey and Halls, 1984], and modal analysis [Van Alstine,1980]. Even with such analyses, the calculated mean directions still can be influenced by secondary components.

From the analysis of samples from four localities in unit I2, Figure 2a shows the distribution of NRM directions for 89 samples and the endpoint directions following progressive thermal demagnetization analysis to 675°C. The mean direction for the 89 samples following cleaning is given in Table 2. We note that not all of the 89 samples responded satisfactorily to the demagnetization treatment. Evidence for a "streaked" distribution of directions similar to the streak seen in the NRM directions remains. However, for 52 of the samples that have orthogonal demagnetization diagrams similar to the diagram shown on Figure 3a, a more tightly grouped population of directions is obtained that lacks the obvious streak. The direction for this group (Table 2) is considered to represent the best estimate of the mean direction for unit I2. A virtually identical direction has been obtained from principal component analysis of 17 samples that have demagnetization diagrams similar to diagrams shown on Figures 3b and 3c (Table 2). The groupings of directions employing 52 samples and 17 samples are very tight and support an inference that these identical directions reflect a very stable, ancient, and possibly original direction of magnetization. The unblocking temperatures for the mineral carrying the characteristic component (hematite) range from above 650°C to 675°C, near the Curie Point (usually considered to be ~680°C [see Merrill and McElhinny, 1983, p. 72]). A final cleaning step, conducted at 680°C in an attempt to obtain a better determination of direction from the high-temperature component resulted in intensities of magnetization that fell below the noise level of the cryogenic magnetometer.

Secondary magnetizations in rocks of unit I2 appear to reflect (1) an overprinting in the present field (the low-temperature component) and (2) an additional earlier overprinting (seen for some samples, particularly samples from locality 3; see Figure 4). For these samples the direction of the earlier overprint is generally in the southeast quadrant of the equal-area net (Figure 3d). Although it is tempting to correlate this overprint with the overprint observed in some of the younger, early Paleozoic rocks (units C02, C07, and C08), evidence within the I2 group of samples alone is inadequate to establish either the direction or the time of acquisition of this prepresent field overprint.

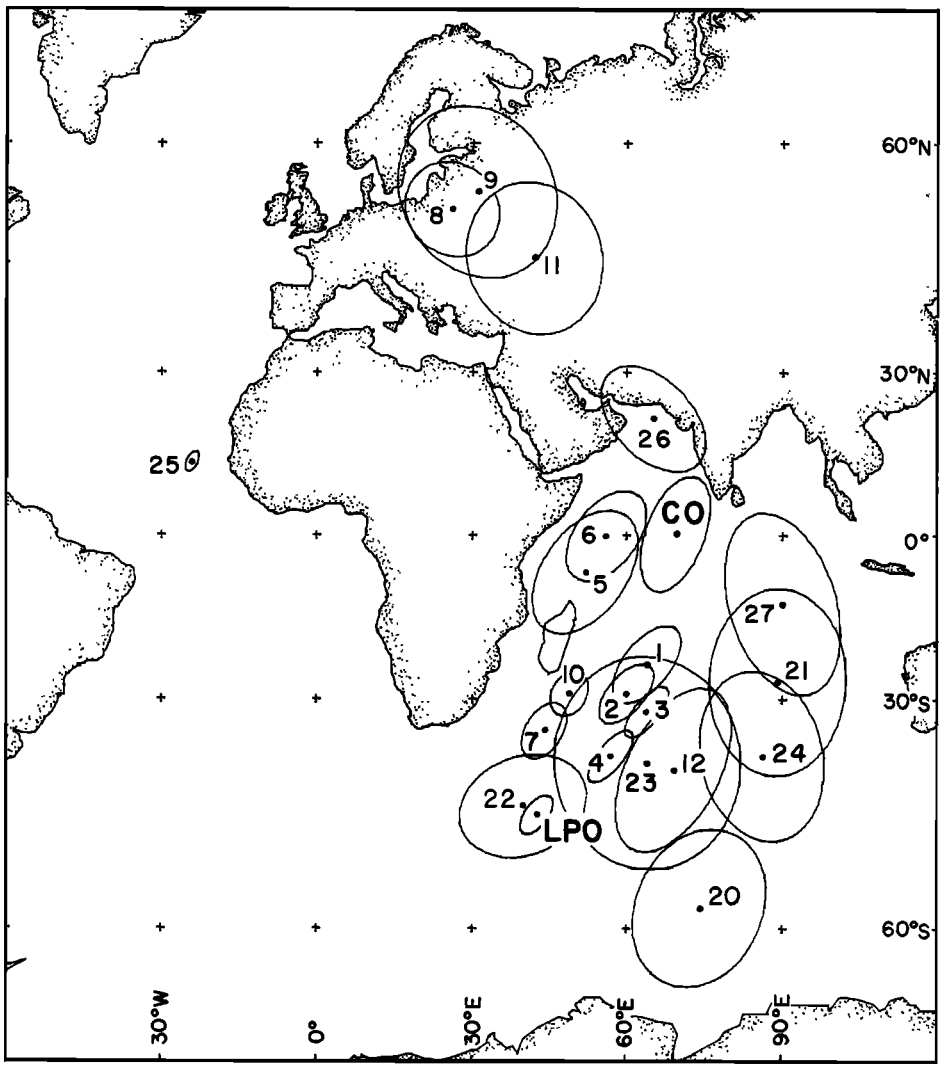


Fig. 7. Plot of paleomagnetic poles from Paleozoic rocks of Africa (Tables 2 and 3). LPO, late Paleozoic overprint (this study); CO, Cambrian pole (this study).

Unit I9. Nineteen samples were collected in a small area of exposure. One sample was lost during the demagnetization process. Of the remaining 18 samples, three displayed a distinctive low inclination direction (Figure 2b) that is in the northeast quadrant of the equal-area diagram after demagnetization. It is a secondary direction that also is seen in several samples from unit CO1 (Figure 2c). These samples thus were not used for the calculation of a mean direction. For the remaining 15 samples, thermal demagnetization revealed two polarities. Directions of samples from the lower part of the section at locality 1 are in the northwest from the upper part of the section at locality 1, and from locality 2 (Figure 3f), are in the southeast quadrant. Although the two groups of directions are not as tightly clustered as desired, we report a mean direction for unit I9 (Table 2) and note that the averaging of the N and R polarity directions helps reduce any systematic bias arising from unremoved components of secondary magnetization. We also note that additional data are needed to confirm the primary origin of this direction and to improve the mean direction.

Paleozoic Strata

Unit CO1. NRM directions from this locality form a southeast trending streak, but cleaning failed to reveal a well-grouped cluster of directions (Figure 2c). Most of the samples are in the eastern quadrant of the equal area net at the 660°C cleaning step. No mean direction or pole is reported for unit CO1 in Table 2 because the dispersion following attempts at cleaning remained large ($k = 7$, $a_{95} = 25°$).

Unit CO2. NRM directions form a streak between the present field direction and a well-clustered low inclination direction that is in the southeast quadrant following cleaning (Figure 2d). Only one component of magnetization remained after removal of the present field component (Figure 3g). The mean direction for this was calculated by the least squares method between the 300°C step of the thermal demagnetization analysis and the origin (Table 2). This same direction has been found as a secondary component in overlying Paleozoic strata. We thus consider the "cleaned" direction to be a secondary overprint direction that has completely masked an original direction. The time

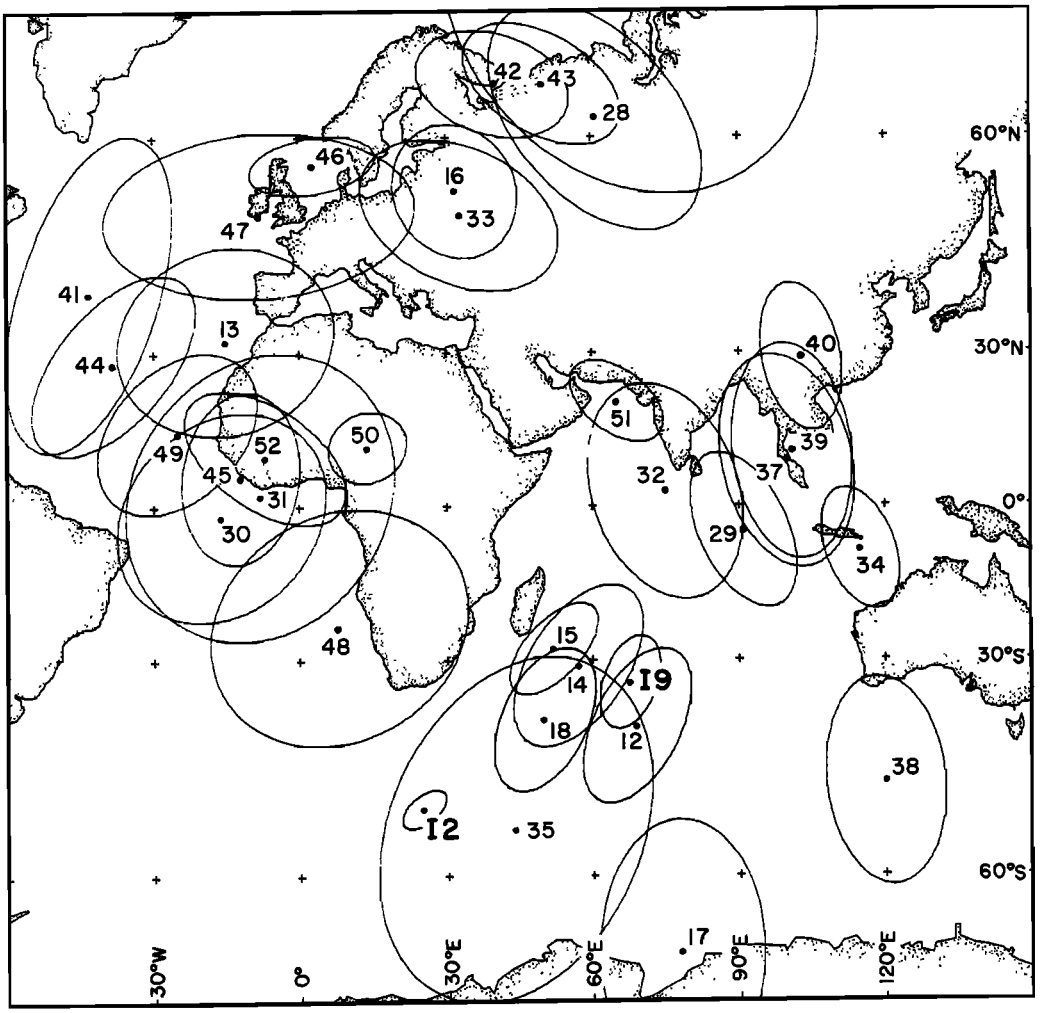


Fig. 8. Plot of paleomagnetic poles from middle and late Proterozoic rocks of Africa (Tables 2 and 3). I9, late Proterozoic pole (this study); I2, middle Proterozoic pole (this study).

of development of this secondary magnetization is discussed later.

Units CO7 and CO8. Two localities were sampled at In Tourfine Pass (Figure 1b). One locality includes the boundary between units CO7 and CO8, and the other lies entirely within unit CO8. Strata from the two localities exhibit mixed polarity. Strata at a third locality in unit CO8, at Amojjiar Pass approximately 45 km to the northeast (Figure 1b), exhibit only a single polarity. Progressive thermal demagnetization analysis revealed multicomponent magnetizations. A component that seems related to the present field direction was first removed by about 300°C. A second component, virtually identical with the secondary direction obtained from unit CO2, also was revealed by the demagnetization analysis. This component was the last one resolved in some samples, whereas in other samples it was resolved in the interval from about 300°C to 600°C. The mean direction for this secondary magnetization (D = 145.7°, I = 11.8°, k = 41.1, a_{95} = 3.0°) was calculated from principal component analysis of 50 samples, including all samples from unit CO2 (Table 2 shows the mean directions obtained from each locality).

For 23 samples from units CO7 and CO8, a third component exhibiting mixed polarity was resolved between 600° and 675°C (Figures 3h and 3i). Eight of the samples provided a mean direction of D = 260.8°, I = 5.5° (k = 11.9, a_{95} = 16.8°), and the remaining 15 samples provided a mean direction of D = 99.1°, I = 24.6° (k = 13.0, a_{95} = 11.0°). The directions are not closely antiparallel, indicating that these "cleaned" directions are still slightly contaminated by overprinting. Nonetheless, the mixed polarity has allowed us to calculate an average direction in which the effects of unremoved components of secondary magnetization are reduced. We believe that this direction reasonably represents the characteristic and primary direction for units CO7 and CO8 (Table 2).

Unit CO10. This unit is a medium- to coarse-grained sandstone. Abundant evidence exists at the outcrop for the past movement of solutions and the widespread precipitation of secondary hematite. Sampling was carried out to investigate the character of the secondary magnetization. No coherent direction of magnetization, either at the NRM step or during cleaning, was obtained from the suite of samples. The directions are highly scattered ($1 < k < 2$), and most samples exhibit very stable, single components of magnetization

TABLE 3. Summary of Selected Paleomagnetic Directions and

No.	Rock Unit	Age of Rock*	Locality	Site Latitude, deg	Site Longitude, deg	N	D, deg
							West
1	Djebel Tarhat red beds	PL	Morocco	33	353	30	121
2	Abadla Series	PL	Morocco	31	357	48	129
3	Choughrane red beds	PL	Morocco	33	354	45	127
4	Taztot trachyandesites	PL	Morocco	32	354	12	136
5	Oued Draa Aftez Limestone Visean	CL	Morocco	29	351	12	112
6	Djebel Hadid red beds, Visean	CL	Morocco	35	353	12	106
7	Gneiguira Supergroup	DM	Mauritania	18	348	44	136
8	Hassi Messaud sediments	CB-O	Algeria	32	6	30	27
9	Oujeft Plateaux Group	CB-O	Mauritania	20	347	7	33
10	Oujeft Plateaux Group, secondary magnetization	CB-O	Maurotamoa	20	347	14	127
11	Volcanics and sediments 532 Ma	CB	Morocco	30	353	64	52
12	Ait Baha Tuffs 570-600 Ma	PCB	Morocco	30	353	87	131
13	Pan African Diorite 616 Ma	Pr	Mali	18	1	62	317
14	Bou Azzer lavas 620-1000 Ma	Pr	Morocco	30	353	43	128
15	Anti-Atlas rocks	Pr	Morocco	31	353	95	129
16	Groupe de Char, 1019 Ma	Pr	Mauritania	20	347	23	34
17	Groupe de Char, secondary magnetization	Pr	Mauritania	20	347	25	336
18	Red sandstone, intermediate temperature component	Pr Pr	Mauritania	17	348	17	137
							Central and
19	Beaufort Group	PU-TL	S Africa	-32	22	348	-62
20	Ecca Sandstones	PL	Tanzania	-10	34	156	52
21	Ketewaka-Mchuchuma coal fields	PL	Tanzania	-10	34	27	117
22	Galula coal field	C-P?	Tanzania	-9	33	34	172
23	Combined red beds	C-P?	Tanzania	-9	33	61	143
24	Vioolsdrif-Haib overprint, 340 Ma?	CU?	SW Africa	-28	17	8	300
25	Hook intrusives, 500 Ma	CB-O	Zambia	-15	27	10	297
26	Doornpoort Formation, 500-550 Ma	CB	S Africa	-24	17	28	229
27	Mulden group DM, 550-560 Ma	CB	Namibia	-21	20	24	278
28	Nama Group N1	PCB	Namibia	-26	17	75	19
29	Nama Group N2	PCB	Namibia	-26	17	15	87
30	Nama Group N3	PCB	Namibia	-26	17	22	122
31	Sijarira Group	PCB	Rhodesia	-17	28	40	291
32	Cunene Anorthosite complex	PCB	Angola	-15	14	75	259
33	Nosib Group NQ2, 660-670 Ma	Pr	Namibia	-21	20	28	12
34	Nosib Group NQ3 NQ2	Pr	Namibia	-21	20	13	102
35	Otavi Group DC2, 3 618-684 Ma	Pr	Namibia	-21	20	10	336
36	Hbozi Complex, 743 Ma	Pr	Tanzania	-9	33	19	10
37	Mbala Dolerites, 678-940 Ma	Pr	Zambia	-9	31	16	84
38	Abercorn Sandstones 678-940 Ma	Pr	Zambia	-9	31	5	318
39	Bukoban intrusives, 803-806 Ma	Pr	Tanzania	-3	31	76	77
40	Bukoban Lavas, 813 Ma	Pr	Tanzania	-3	30	153	239
41	Bukoban Sandstone	Pr	Tanzania	-3	31	12	131
42	Umkondo Dolerite, 650-1150 Ma	Pr	Rhodesia	-19	33	77	183
43	Waterberg Diabase, 600-1250 Ma	Pr	S Africa	-25	28	90	190

Poles Reported From Proterozoic and Paleozoic Rocks of Africa

I, deg	KD	ED	Paleomagnetic Pole Latitude, deg	Paleomagnetic Pole Longitude, deg	dp	dm	Reference
Africa							
4	35	8	-24	64	4	8	Martin et al. [1978]
11	59	6	-29	60	3	6	Morel et al. [1981]
-6	20	5	-32	64	2	5	Daly and Pozzi [1976]
-5	80	5	-39	57	2	5	Daly and Pozzi [1976]
37	80	11	-7	52	8	13	Martin et al. [1978]
39	26	9	0	56	6	10	Martin et al. [1978]
27	89	5	-35	44	3	5	Kent et al. [1984]
77	3		53	26	5	6	Bucur [1971]
61	87	7	55	31	8	11	Morris and Carmichael [1978]
23	89	4	-29	49	3	4	Morris and Carmichael [1978]
66	182	7	47	42	9	11	Daly and Pozzi [1977]
-20		13	-41	69	7	14	Hailwood [1972]
79		9	32	345	16	17	Morel [1978]
10		17	-31	57	9	17	Hailwood [1972]
21	39	9	-28	52	5	10	Rhodes [1976]
61	160	6	54	32	7	9	Morris and Carmichael [1978]
37	36	11	67	258	7	13	Morris and Carmichael [1978]
14	49	13	-40	50	7	13	Kent et al. [1984]
Southern Africa							
			76	240			Ballard and VanderVoo [1985]
	7		-58	74	7	9	Nairn [1964]
54	60	12	-27	89	12	17	Opdyke [1964]
69	227	5	-46	40	7	9	Opdyke [1964]
66			-40	64	16	16	Opdyke [1964]
-52	25	10	39	266	9	14	Hargraves and Onstott [1985]
51			14	336	1	2	Brock [1967]
-43	58	9	-22	245	7	11	Piper [1975a]
-34	18	16	13	270	10	18	McWilliams and Kroner [1981]
-11	27	15	62	61	7	15	Kroner et al. [1980]
31	41	15	-5	91	9	17	Kroner et al. [1980]
-67	25	14	2	164	19	23	Kroner et al. [1980]
67	9	18	2	352	26	30	Reid [1968]
-46	7	17	-3	255	15	22	Piper [1974]
36	26	12	51	33	8	14	McWilliams and Kroner [1981]
-13	11	13	-9	115	7	13	McWilliams and Kroner [1981]
-65	11	15	55	224	19	24	McWilliams and Kroner [1981]
11	19	18	72	68	9	18	Piper [1975b]
40	19	18	9	100	13	22	Piper [1975b]
-14	26	15	49	300	8	16	Piper [1975b]
32	5	19	11	101	12	22	Piper [1972]
-25	5	13	-29	283	7	14	Piper [1972]
-19	7	20	-40	137	11	21	Piper [1972]
-10	50	7	-65	220	4	7	McElhinny and Opdyke [1964]
2	31	8	-65	230	4	8	Jones and McElhinney [1966]

TABLE 3.

No.	Rock Unit	Age of Rock*	Locality	Site Latitude, deg	Site Longitude, deg	N	D, deg
						Central and Southern	
44	Nosib Group NQ1, 600-1250 Ma	Pr	Namibia	-21	20	25	312
45	Kisii Series, 906-1200 Ma	Pr	Kenya	-1	35	63	99
46	Koras Group, 1000-1200 Ma	Pr	S Africa	-28	21	7	170
47	Nosib Group, NBX	Pr	Namibia	-24	16	11	345
48	Central Namaqua belt, 1050-1150 Ma	Pr	SW Africa	-30	20		304
49	Ookiep basic intrusion, 1070 Ma	Pr	S Africa	-30	18	22	132
50	Van Dyk Dolerite Dyke, 1120 Ma	Pr	S Africa	-26	28	8	158
51	National Kimberlite, 1140 Ma	Pr	S Africa	-26	28	11	221
52	Harper amphibolite, 1.9-2 Ga	Pr	Liberia	7	351	11	281

*Ages: PU-TL, Permian-Triassic; PL, Lower Permian; C-P, Carboniferous-Permian; CU, CB-O, Cambrian-Ordovician; CB, Cambrian; PCB, Proterozoic-Cambrian; Pr, Proterozoic

(Figures 2f, 3j, and 3k). Thus no mean direction of magnetization can be reported.

Discussion

Mean directions determined for the individual units are shown in Figure 2. Units I2, I9, C07, and C08 appear to contain primary as well as secondary magnetizations. Unit C02, in contrast, contains only secondary magnetizations, the mean direction of which is shown in Figure 5. We report four poles from Mauritania (Table 2) and provisionally assign them to the Middle Proterozoic, Late Proterozoic, Cambrian, and late Paleozoic (probably Permian). The Late Proterozoic and Cambrian poles come from strata containing mixed polarity.

Before discussing paleomagnetic poles calculated from the mean directions reported in Table 2, we first compare some of the directions with results reported from an earlier study of the same units in the Adrar de Mauritanie. Paleomagnetic directions reported by Morris and Carmichael [1978] for units I2, C08, and C010 are also shown in Figure 6. Additionally, the direction reported by Kent et al. [1984] for a unit correlative with unit C010, which they sampled about 300 km south of Atar, also is shown on Figure 6. Kent et al. [1984] pointed out that Morris and Carmichael's [1978] "initial" directions for units I2, C08, and C010 lie close to a great circle (short dashed lines, Figure 6) that connects Morris and Carmichael's "secondary" direction for units C08 and C010 (dot 6, Figure 6) with the present field direction (cross of Figure 6). A great circle drawn from our mean late Paleozoic overprint direction (dot 8, Figure 6) through the present field direction (long dashed line, Figure 6) also passes through the "initial" directions that Morris and Carmichael report for units I2, C08, and C010 (dots 3, 5, and 4, Figure 6). We also observed approximately the same "intermediate" direction in some samples of unit I2 prior to their cleaning at high temperatures. We therefore suggest that these directions result from some combination of unremoved secondary components of magnetization. Last, we note that Morris and Carmichael's as well as Kent et al's secondary directions (dots 6 and 7, Figure 6) fall near a great circle that connects our late Paleozoic overprint direction with our Cambrian direction (solid lines connecting dots 8 and 10, Figure 6), and that directions 6 and 7 are slightly offset toward the Cambrian direction. The Cambrian rocks of Mauritania studied by Morris and Carmichael and Kent et al., although heavily overprinted by a much later magnetization, thus may still contain components of an original Cambrian magnetization.

Figures 7 and 8 depict selected poles reported from Paleozoic and Proterozoic strata elsewhere in Africa (Table 3). Poles summarized in Table 3 were derived from compilations of Irving [1960a, b, 1961, 1962a, b, 1963, 1965] and McElhinny [1968a, b, 1969, 1970, 1972a, b, 1977, 1978, 1980], and from a compilation of ours for the publication interval 1978 to present.

Paleomagnetic criteria used to select poles for the compilation (Table 3) were (1) demagnetization analysis, (2) number of samples => 7, (3) a_{95} =< 20, and (4) kappa => 5. Use of these criteria reduced the number of candidate poles from 39 to 21 for Paleozoic rocks and from 96 to 71 for Proterozoic rocks. An additional reduction in the number of potentially correlative poles from Proterozoic rocks, from 71 to 31, was effected by elimination of poles from rocks having apparent ages greater than early middle Proterozoic. The data in Table 3 are assembled in two groups: (1) West Africa and (2) Central and Southern Africa. Within each group, poles are listed in approximate temporal order.

Late Paleozoic Overprinting

The secondary direction isolated in units C02, C07, and C08 (Figure 5 and Table 2) appears to be

(continued)

I, deg	KD	ED	Paleomagnetic Pole Latitude, deg	Longitude, deg	dp	dm	Reference
Africa (continued)							
40	21	15	28	322	10	18	McWilliams and Kroner [1981]
-59	24	11	-6	168	12	17	Brock et al.[1972]
-7	67	7	-57	183	3	7	Briden et al.[1979]
24	8	18	51	352	10	19	Kroner et al. [1980]
84	37	11	-24	8	21	22	Peacock et al. [1985]
-48	28	15	-15	155	13	19	Piper [1975b]
-66	140	5	-12	194	7	8	Jones and McElhinney [1966]
-51	41	7	-21	245	6	10	Hargraves and Onstott [1980]
-13	44	19	10	253	10	19	Onstott et al. [1984]

Upper Carboniferous; CL, Lower Carboniferous; DU, Upper Devonian; DM, Middle Devonian;

present in unit CO1 and perhaps also in units I2 and I9. Similar directions have been reported from various Paleozoic strata elsewhere in West Africa (Morris and Carmichael [1978] (Table 3, pole 10); Kent et al. [1984] (Table 3, pole 18)). Such directions are now becoming generally recognized as reflecting a regionally pervasive remagnetization that occurred sometime near the end of the Paleozoic Era. Pole LPO (Table 2) is in the vicinity of Permian poles reported from Africa (Figure 7 and Table 3; poles 1-4 and 22, 23). Last, two poles that have been reported as Devonian (Figure 7 and Table 3; pole 7) and as Cambrian (Figure 7 and Table 3; pole 12), and some late Proterozoic poles as well (Figure 8), also are in the same general area. This correspondence supports the idea of a general late Paleozoic overprinting for Africa, similar to overprinting observed in Europe as well as in North America. This overprinting may have been impressed on the rocks by chemical processes [McCabe et al., 1983], or by a Vp TRM [Kent, 1985]. However, relating this overprint to a specific geologic event has remained difficult, and the cause for the pervasive overprinting remains obscure.

Early Paleozoic Pole

Isotopic ages and paleontologic data from below and above units CO7 and CO8 serve to place these strata in the Cambrian Period (Figure 1). Thirteen Cambrian and late Proterozoic pole positions are listed in Table 3. They include poles that have ages reported to lie near the late Proterozoic-Cambrian boundary (PCB, Table 3) and near the Cambrian-Ordovician boundary (CB-O, Table 3). Four of these broadly correlate with the normal and reverse polarity directions (and poles) from units CO7 and CO8 (Figures 7 and 8, and Table 3; poles 26, 27, 29, and 32). However, four other poles (Figures 7 and 8, and Table 3; poles 8, 9, 11 and 28) group in a different, higher latitude position. A third, still different group (Figures 7 and 8; poles 25, 30, and 31) also is present. Lastly, two poles (Figure 7, Table 3; poles 10 and 12) have positions that may reflect the late Paleozoic overprinting. The scatter in reported pole positions for the Cambrian of Africa could arise from one or more of the following reasons: remagnetization, insufficient cleaning or the failure of cleaning to completely remove secondary components of magnetization, and problems with age control.

Middle and Late Proterozoic Poles

Middle and late Proterozoic poles selected for Africa are shown in Figure 8. We also have included poles for rocks that have ages reported as falling close to the Proterozoic-Cambrian boundary (PCB in Table 3). We note that the scattered distribution of poles shows no evident correlation with ages reported for the rocks.

The middle Proterozoic pole we report from unit I2 has no correlative in the African paleomagnetic data set. Poles reported for the same general age (Figure 8 and Table 3, poles 42-51) display very different positions. No correlations are possible with the existing data set.

The paleomagnetic directions we report from unit I9, which we consider to be Late Proterozoic, display two polarities. The south seeking pole position is close to some other reported Late Proterozoic poles (Figure 8 and Table 3, poles 12, 14, 15, and 18). This grouping plots in the area of poles reported for the late Paleozoic of Africa (from Permian rocks and from older rocks apparently overprinted in the Permian). It is a coincidence that immediately raises a question as to the primary nature of the late Proterozoic direction that we report from unit I9. Unit I9, however, contains both polarities whereas only a single (reverse) polarity has been observed in directions attributable to the late Paleozoic remagnetization in Africa, which supports the idea of an early magnetization in strata of unit I9.

Nonetheless, pole positions for the Proterozoic of Africa are too scattered to suggest a correlation, and the determination of our Late Proterozoic pole is not well enough established to report the magnetization in unit I9 as an unquestioned primary magnetization.

Conclusions

The distributions of poles shown in Figures 7 and 8 depict problems that are faced when poles are to be selected for the construction of a hoped-for reliable polar path for the Proterozoic and Paleozoic of Africa. A number of solutions are possible, as demonstrated by differing polar paths that already have been proposed by a number of workers [e.g., Morel and Irving, 1978; Kroner et al., 1980; Van der Voo et al., 1980; McWilliams, 1981; Piper, 1982]. The reversed polarity pole position for unit CO8 is in good agreement with all of the proposed paths except for the path reported by Morel and Irving [1978], but even in this case we note that the paleolatitude of pole CO8, which is about 7°, fits well with their proposed paleocontinental map for 600 Ma. Additionally, the fact that Paleozoic control points for Africa are widely separated in time results in a polarity dilemma. The paths proposed by Kroner et al. and by McWilliams for Cambrian time have reversed senses of motion. We suggest, therefore, that data are as yet inadequate to allow a valid polar path to be constructed for the Paleozoic and Proterozoic of Africa.

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